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	School of Digital Technologies
	ITM-390 Machine Learning

Final Project Report

Project Information			
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I. Introduction

Managing personal finances has become increasingly complex in the modern era. The widespread adoption of digital payment platforms, online shopping services, subscription-based applications, and multiple income streams has made it difficult for individuals — especially university students — to effectively monitor their spending habits and plan their budgets. Despite the availability of various budgeting applications, most rely on manual data entry and provide only static visualizations with limited analytical depth. These tools lack intelligent automation, predictive capabilities, and personalized financial feedback, which significantly reduces their practical value for users seeking meaningful insights into their financial behavior.

Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML) have opened new possibilities for automating and enhancing personal finance systems. Prior research demonstrates that supervised learning models such as Logistic Regression and TF-IDF-based text classifiers can accurately classify financial transactions, while time-series forecasting methods such as rolling average and weighted recent models can reliably predict future spending trends based on personal historical data [1][2]. Despite these capabilities, many existing financial AI systems are designed for large-scale banking environments, involve complex architectures that are difficult to interpret, and are not accessible for student-level applications [3].

This project proposes an AI Budget Tracker with Smart Financial Insights — a three-model AI pipeline designed to address these limitations. The system integrates three distinct AI paradigms:

- A supervised text classification model (TF-IDF + Logistic Regression) for automatic expense categorization
- A personalized on-demand spending forecasting model using rolling average and weighted recent algorithms with walk-forward validation, replacing the initially

proposed linear regression approach which performed poorly on irregular spending data

- A rule-based inference engine for generating human-readable financial insights and overspending warnings

The primary dataset for classification is sourced from the Personal Expense Classification Dataset on Kaggle [4], while the prediction model trains directly on each user's own historical transaction data, making it fully personalized and requiring no external dataset. This integrated approach demonstrates how multiple AI techniques can be applied within a single, interpretable, and academically relevant framework to support smarter financial decision-making for students in Cambodia.

II. Literature review

Artificial Intelligence and Machine Learning have been extensively applied to personal and corporate financial systems. This section reviews prior work in three key areas directly relevant to the proposed system: transaction classification, spending prediction, and rule-based financial insight generation.

A. Expense Classification Using Machine Learning

Transaction classification is a well-studied problem in financial AI. Supervised learning models — including Naive Bayes, Logistic Regression, and Support Vector Machines (SVM) — have been widely used to assign textual transaction descriptions to predefined spending categories such as food, transportation, and entertainment. Siebert et al. [1] demonstrated that Naive Bayes classifiers achieve strong baseline performance on short, noisy financial text due to their robustness with limited training data. More recent work by Golbayani et al. [2] showed that combining keyword-based rule systems with lightweight supervised classifiers can outperform purely data-driven approaches when labeled datasets are small, as is typically the case in personal finance contexts. This hybrid strategy directly informs the design of Model 1 in this project.

B. Spending Prediction Using Regression and Time-Series Models

Predicting future expenditure from historical transaction data is a core challenge in personal finance systems. Linear Regression has been shown to be an effective baseline for capturing spending trends when data is aggregated at the monthly level, as demonstrated by James et al. [3], who establishes regression as a fundamental tool for modeling financial behavior. More advanced approaches such as ARIMA and Long Short-Term Memory (LSTM) neural networks, as described by Chollet [6], have demonstrated superior performance for highly non-linear and seasonal spending patterns. However, these methods require substantially more data and computational resources than are available in a student-level academic context. Moving average and weighted, recent forecasting methods have been shown to be effective lightweight alternatives when historical data is limited, providing reasonable prediction accuracy without requiring a large pre-collected dataset [3]. Given the small dataset size of this project — six months of personal transaction data — a personalized on-demand rolling average forecasting approach was selected, with LSTM identified as a direction for future work.

C. Rule-Based Inference and Expert Systems in Finance

Rule-based expert systems represent one of the earliest paradigms in AI and continue to be valuable in financial advisory contexts due to their transparency and interpretability. Heaton [4] discusses the application of knowledge-based systems to personal finance, highlighting their effectiveness in generating explainable recommendations based on predefined thresholds — for example, alerting users when spending in a specific category exceeds a percentage of their income. Unlike black-box machine learning models, rule-based systems allow users to understand and trust the generated advice. This property is particularly important for a student-facing application where clarity and explainability are prioritized over predictive complexity. Model 3 of this project applies this approach through a custom inference engine built on standard budgeting guidelines.

D. Gap Identified

While prior studies have individually addressed classification, prediction, and financial advisory tasks, few systems integrate all three paradigms within a single, lightweight, and interpretable pipeline designed for student-level financial analysis. Most existing AI financial tools are either too complex for academic use or are designed for institutional banking contexts with access to large proprietary datasets [5]. This project addresses that gap by combining rule-

based AI, supervised regression, and expert-system inference within a unified, open-source, and academically deployable framework.

III. Methodology

This project adopts a machine learning-based research methodology consisting of three interconnected AI models, each targeting a specific financial analysis task. The overall pipeline follows a sequential flow: raw user transaction data is first classified by Model 1, aggregated data is then passed to Model 2 for spending prediction, and the combined outputs are analyzed by Model 3 to generate actionable financial insights.

A. System Workflow

The system pipeline proceeds through the following steps:

Figure 1: System Architecture of AI Budget Tracker System

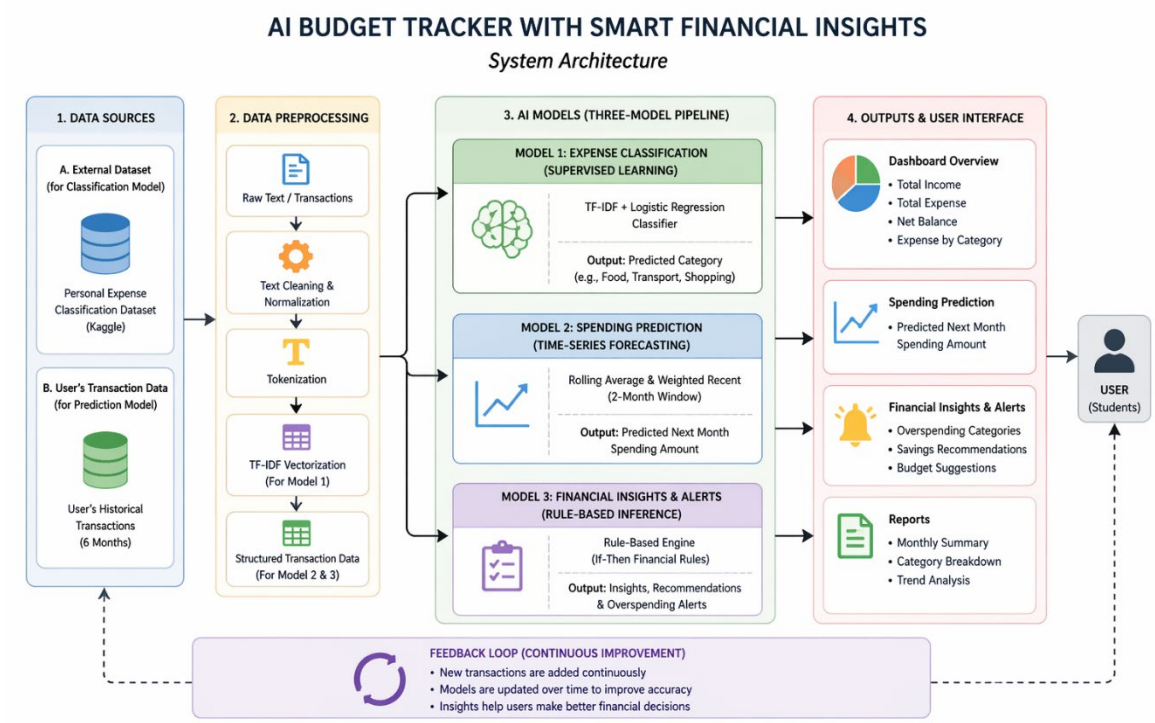


Figure 1: *System Architecture of AI Budget Tracker with Smart Financial Insights*

The overall system architecture is illustrated in Figure 1.

- User inputs income and expense records into the system
- Model 1 processes each expense description and assigns a category (e.g., Food, Transportation)
- Categorized data is aggregated by time period (monthly) for Model 2
- Model 2 applies on-demand per-user forecasting to predict the next month's total expenditure
- Model 3 receives category totals, income, and Model 2 predictions as inputs
- Model 3 evaluates spending against predefined budget thresholds and generates text-based insights
- Results (categorized expenses, prediction, insights) are displayed to the user via tables and charts

B. Model 1 — Expense Classification Model

Objective: Automatically classify each expense transaction into one of six predefined categories: Food, Transportation, Entertainment, Utilities, Rent, or Other, based on the textual description of the transaction.

AI Type: Supervised Machine Learning — TF-IDF Vectorizer with Logistic Regression.

Input / Output:

- Input: Transaction description (plain text string)
- Output: Expense category label

Training Method:

Model 1 was trained in a dedicated training notebook on the Kaggle dataset and saved as a reusable artifact (tfidf_expense_classifier.joblib). The pipeline loads this pre-trained artifact at runtime for inference. The model consists of a TF-IDF vectorizer that converts transaction descriptions into numerical feature vectors, combined with a Logistic Regression classifier that predicts the expense category. Numerical features such as transaction amount, day of week, month, and weekend flag are scaled and appended to the text features before classification. A clean text preprocessing function normalizes the input by lowercasing, removing punctuation, and standardizing common terms (e.g., "gasoline", "petrol", "fuel" are all normalized to "petrol"). A category mapping layer (CATEGORY_MAP) then

converts the model's raw output labels into the six proposal-defined categories. Example mappings include:

- Shopping, Healthcare, Education, Misc → Other
- Housing → Rent

Dataset:

Personal Expense Classification Dataset (Kaggle, S. Deseker [4]). This dataset contains labeled transaction descriptions with corresponding category annotations, used to train and evaluate the TF-IDF + Logistic Regression classifier.

C. Model 2 — Spending Prediction Model

Objective: Predict the total spending amount for the next month based on the user's own historical monthly expense data.

AI Type: Personalized On-Demand Time-Series Forecasting — Rolling Average and Weighted Recent algorithms with walk-forward validation.

Input / Output:

- Input: User's historical monthly expense totals (aggregated from Model 1 output)
- Output: Predicted total expense for the next month (USD) and spending trend (Upward / Downward / Stable)

Training Method:

Linear Regression was initially considered but removed during implementation as it learns a global trend and performs poorly on spiky or irregular spending patterns. Rolling Average and Weighted Recent algorithms were selected as more adaptive replacements. Unlike a static pre-trained model, Model 2 trains fresh on each individual user's own transaction history at runtime — no external dataset is required. Monthly expense totals are first aggregated from the classified transaction data. Walk-forward validation is then performed across window sizes of 2 to 5 months, testing both Rolling Average and Weighted Recent algorithms on each window. The combination with the lowest Mean Absolute Error (MAE) is automatically selected as the best model for that user. The final forecast is generated using the selected algorithm and window. Spending trend is determined by comparing the most recent expense value against the average of the best window — an increase above $20 is

classified as Upward, a decrease below -$20 as Downward, and anything in between as Stable.

Dataset: No external dataset required for production use. For evaluation purposes, a 36-month historical dataset (January 2022 – December 2024) with realistic monthly spending patterns and seasonal December spikes was used to properly assess walk-forward validation performance across window sizes.

D. Model 3 — Financial Insight and Overspending Detection Model

Objective: Convert numerical financial data — income, category-level expenses, and predicted spending — into clear, human-readable financial insights, overspending warnings, and budget recommendations.

AI Type: Rule-Based Inference System (Expert System).

Input / Output:

- Input: Total income, per-category expense totals, predefined budget thresholds, and Model 2 prediction
- Output: Textual financial insights

Inference Rules: The following budget rules were defined based on standard personal finance guidelines:

- Food ≤ 25% of income
- Transportation ≤ 15% of income
- Entertainment ≤ 10% of income
- Utilities ≤ 10% of income
- Rent ≤ 30% of income
- Savings ≥ 20% of income

Each rule is evaluated against actual spending ratios derived from Model 1's categorized output. The engine produces three status levels: OVERSPENDING (exceeds threshold), WARNING (exceeds 85% of threshold), and OK (within budget). Violations trigger a corresponding NLP template to generate a readable warning. For example, if food spending equals 31.6% of income, the engine produces: "You spent 31.6% of your income on Food, which exceeds the recommended limit of 25%. Consider reducing Food by $89.73."

E. System Interface — Telegram Bot

The system was deployed with a Telegram bot as the front-end interface for the demonstration. Users interact with the pipeline by uploading their personal transaction CSV file directly to the bot. The bot then runs the complete three-model pipeline sequentially:

- Model 1 classifies all expense transactions by category
- Model 2 generates the next month's spending forecast
- Model 3 evaluates spending against budget thresholds and generates insights

The full Smart Financial Insights Report is returned to the user as a formatted text message inside the Telegram chat. This interface was selected for three reasons:

- No additional app installation required
- Accessible on both mobile and desktop
- Widely used in Cambodia making it familiar to the target student users

IV. Experiments

A. Experimental Setup

All experiments were conducted on a local development machine running Windows 11, Python 3.12, and the following key libraries: Pandas, NumPy, Scikit-learn 1.8, Matplotlib, Scipy, and Joblib.

Model 1 was trained separately in a dedicated training notebook and exported as a reusable artifact. Models 2 and 3 were run directly inside the main pipeline notebook. The full pipeline demonstration was executed in Jupyter Notebook using VS Code.

B. Model 1 Experiment — Expense Classification

Dataset: The Personal Expense Classification Dataset was sourced from Kaggle via HuggingFace, containing 51,482 labeled transaction records across 15 raw categories including Food, Transport, Entertainment, Utilities, Housing, Shopping, Healthcare, Education, Misc, Childcare, Investment, Freelance, Other Income, Salary, and Rental. Only expense records were used for classification training. The dataset was split 80/20 with random_state=42.

Setup: Two approaches were trained and compared to select the best classifier:

Approach 1 — Keyword-Based Baseline: A simple keyword rule dictionary was manually constructed for four categories: Entertainment, Education, Healthcare, and Shopping. Each transaction description was cleaned using the **clean_text()** function and matched against the keyword list. Transactions with no keyword match were labeled "Unknown."

Approach 2 — TF-IDF + Logistic Regression: The following steps were applied:

- clean_text() preprocessing — lowercasing, punctuation removal, and fuel term normalization (gasoline/petrol/diesel → petrol)
- TF-IDF vectorization of cleaned transaction descriptions
- Numerical feature extraction — Amount, Type, Month, Day of Week, Is Weekend
- StandardScaler applied to numerical features
- Sparse feature matrix constructed by combining TF-IDF vectors and scaled numerical features using **hstack()**
- Logistic Regression trained with max_iter=1000, solver=lbfgs, C=1.0

Key modification from proposal: The proposal originally described a rule-based keyword dictionary as the primary classification method. During implementation, this was replaced with a pre-trained TF-IDF + Logistic Regression model after the keyword baseline demonstrated insufficient accuracy at 78.71%. The keyword approach was retained only as a comparison baseline.

C. Model 2 Experiment — Spending Prediction

Dataset:

A 36-month historical dataset (January 2022 – December 2024) was used containing realistic monthly spending patterns with a gradual upward trend and recurring December seasonal spikes ($1,800 in December 2022, $2,200 in December 2023, $2,500 in December 2024). This larger dataset was specifically chosen to enable meaningful walk-forward validation across window sizes 2 to 5 months since the 6-month demo dataset was too small for reliable evaluation.

Setup:

Monthly expense totals were used as the input time series. Walk-forward validation was applied as follows:

- For each window size W (2, 3, 4, 5 months), all months before time T were used to predict month T
- This was repeated for every valid T in the dataset ensuring no data leakage
- Both Rolling Average and Weighted Recent algorithms were tested for each window size
- MAE was computed for each algorithm and window combination
- The combination with the lowest MAE was automatically selected as the best model

Key modification from proposal: Linear Regression was initially proposed as the prediction method. During implementation it was removed after observing that the dataset contained irregular spending spikes that a global linear trend could not capture accurately. Rolling Average and Weighted Recent were selected as more adaptive replacements that focus only on the most recent N months rather than fitting a global trend across all data points.

D. Model 3 Experiment — Financial Insight Generation

Dataset: Model 3 does not require its own dataset. It receives all inputs directly from the outputs of Model 1 and Model 2 at runtime:

- Category expense totals — from Model 1's classified transactions
- Total income and net balance — from the user's raw CSV
- Next month forecast and trend — from Model 2's prediction

Setup: The rule-based inference engine evaluated each spending category against the predefined budget thresholds. For each category the engine:

- Calculated spending as a percentage of total income
- Compared against the defined threshold
- Assigned OVERSPENDING, WARNING, or OK status
- Generated a readable NLP text insight using the template system
- Produced a full Smart Financial Insights Report

The pipeline demo dataset (January 2024 – June 2024, monthly income $500.00) was used. Model 3 evaluated the latest month (June 2024) using category totals from Model 1 and the $136.35 Stable forecast from Model 2.

V. Results

Since this system is designed as an on-demand per-user pipeline, each model operates independently on the data provided by the user at runtime. For demonstration purposes, Model 1 was evaluated using a dedicated training notebook on the full Kaggle dataset containing 51,482 records. Models 1 and 3 were demonstrated using a six-month personal transaction demo dataset (January 2024 – June 2024, monthly income $500.00). Model 2 was evaluated on a separate 36-month historical dataset (January 2022 – December 2024) to enable meaningful walk-forward validation. Model 3 was additionally demonstrated through the Telegram bot interface using two CSV dataset. This separation reflects the real-world design of the system where users may provide varying amounts of historical data.

A. Model 1 — Expense Classification Results

Model 1 successfully classified all 36 expense transactions from the demo dataset into the six predefined categories using the pre-trained TF-IDF + Logistic Regression classifier loaded from tfidf_expense_classifier.joblib. The classifier achieved an overall accuracy of 99.86% on the held-out test set, significantly outperforming the keyword-based baseline of 78.71%.

Figure 2: presents the model accuracy comparison between the two approaches tested during training.

	Model	Accuracy
0	Keyword-Based	0.787129
1	TF-IDF + Structured	0.998640
2	MiniLM + Structured	NaN

Figure 3: presents the category distribution produced by Model 1 on the demo dataset across all six months.

```
[5]
...  All expense transactions classified!

     Category distribution (proposal categories):
     category
     Food              12
     Transportation     8
     Utilities          8
     Other              4
     Entertainment      2
     Rent               2
```

	date	description	amount	category_raw	category
1	2024-01-05	Bought rice and vegetables at market	4.50	Food	Food
2	2024-01-08	Grab ride to work	3.00	Transport	Transportation
3	2024-01-12	Netflix subscription	15.99	Utilities	Utilities
4	2024-01-18	Electricity bill	30.00	Utilities	Utilities
5	2024-01-22	Coffee at Brown Coffee	2.50	Food	Food
6	2024-01-28	Bought clothes at AEON Mall	45.00	Shopping	Other
8	2024-02-03	Lunch at local restaurant	6.00	Food	Food
9	2024-02-07	Bus fare to university	1.50	Transport	Transportation
10	2024-02-14	Cinema ticket	8.00	Entertainment	Entertainment
11	2024-02-19	Water bill	12.00	Utilities	Utilities
12	2024-02-23	Snacks at 7-Eleven	2.00	Food	Food
13	2024-02-27	Fuel at Caltex	5.00	Transport	Transportation

category distribution produced by Model 1

Food recorded the highest number of transactions (12), reflecting frequent small daily purchases such as meals and groceries. Rent recorded the highest total amount ($400.00) despite only two transactions, as expected for a fixed monthly expense. Figure 1 shows the full classification output from the pipeline.

B. Model 2 — Spending Prediction Results

Model 2 applied on-demand per-user walk-forward validation across window sizes of 2 to 5 months, comparing Rolling Average and Weighted Recent algorithms on the 36-month historical dataset.

Figure 4: Model 2 — Algorithm Comparison

```
📊 Model Evaluation (Walk-Forward Validation):

  Rolling Average (window=2)
  ────────────────────────────────────────
  MAE  (Mean Absolute Error)      : $173.53
       → on average, off by $174
  RMSE (Root Mean Squared Error)  : $236.08
       → penalises big misses harder than MAE
  MAPE (Mean Abs Percentage Error): 11.2%
       → on average, off by 11.2% of actual spend
  R²   (Explained variance)       : 0.4811
       → weak fit — spiky data is hard to predict

  Weighted Recent (window=2)
  ────────────────────────────────────────
  MAE  (Mean Absolute Error)      : $167.16
       → on average, off by $167
  RMSE (Root Mean Squared Error)  : $236.14
       → penalises big misses harder than MAE
  MAPE (Mean Abs Percentage Error): 10.9%
       → on average, off by 10.9% of actual spend
  R²   (Explained variance)       : 0.4809
       → weak fit — spiky data is hard to predict

🏆 Winner by MAE: Weighted Recent
```

Figure 4 shows the Weighted Recent algorithm with a window of 2 months was automatically selected as the best model, achieving the lowest MAE of $167.16 and MAPE of 10.9%. The walk-forward validation confirmed no data leakage during evaluation. The model forecasts January 2025 spending at $2,333.33 with a recent upward spike trend.

Figure 5 presents the full Model 2 evaluation dashboard showing four panels: actual vs predicted spending over time, absolute error per month with MAE reference lines, residuals plot, and MAE vs window size comparison.

Figure 5**:** Model 2 — Monthly Financial Summary (Demo Dataset)

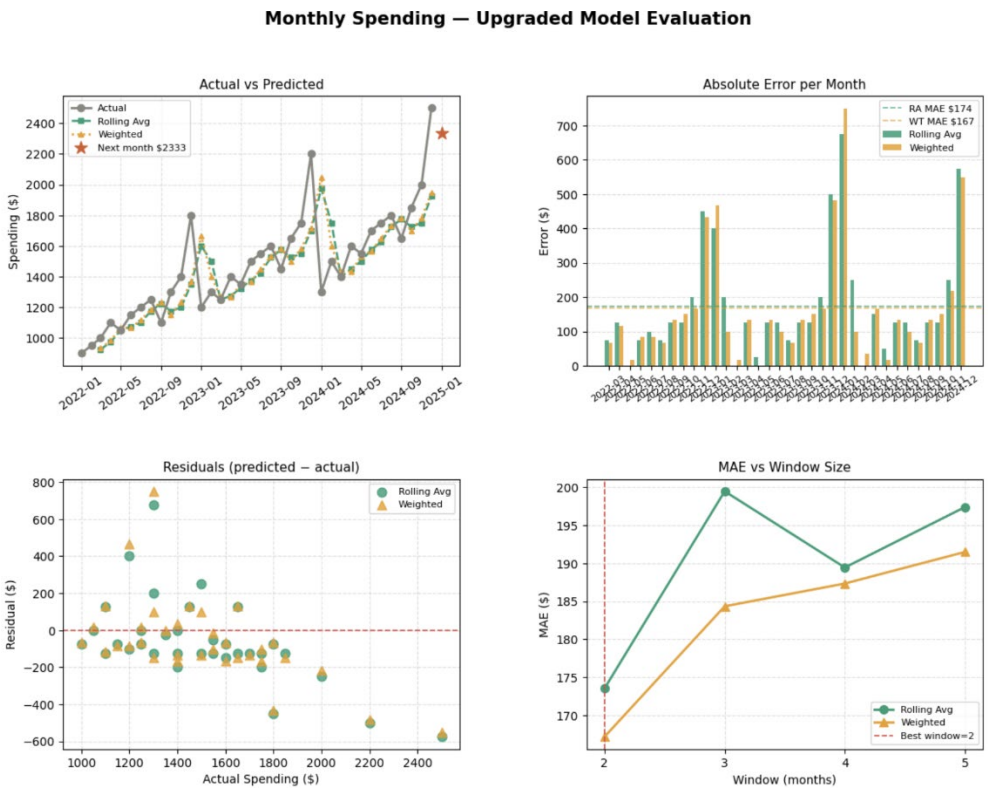


Figure 5 presents the full Model 2 evaluation dashboard showing four panels: actual vs predicted spending over time, absolute error per month with MAE reference lines, residuals plot, and MAE vs window size comparison.

C. Model 3 — Financial Insights Results

Model 3 was demonstrated through the Telegram bot interface using two CSV datasets with a monthly income of $1,350.00. The inference engine evaluated all six spending categories against the predefined budget thresholds and generated a full Smart Financial Insights Report delivered as a formatted Telegram message. Below shows the full Telegram bot process and output.

- Figure 4: Telegram Bot Interface – Classification and Logging.

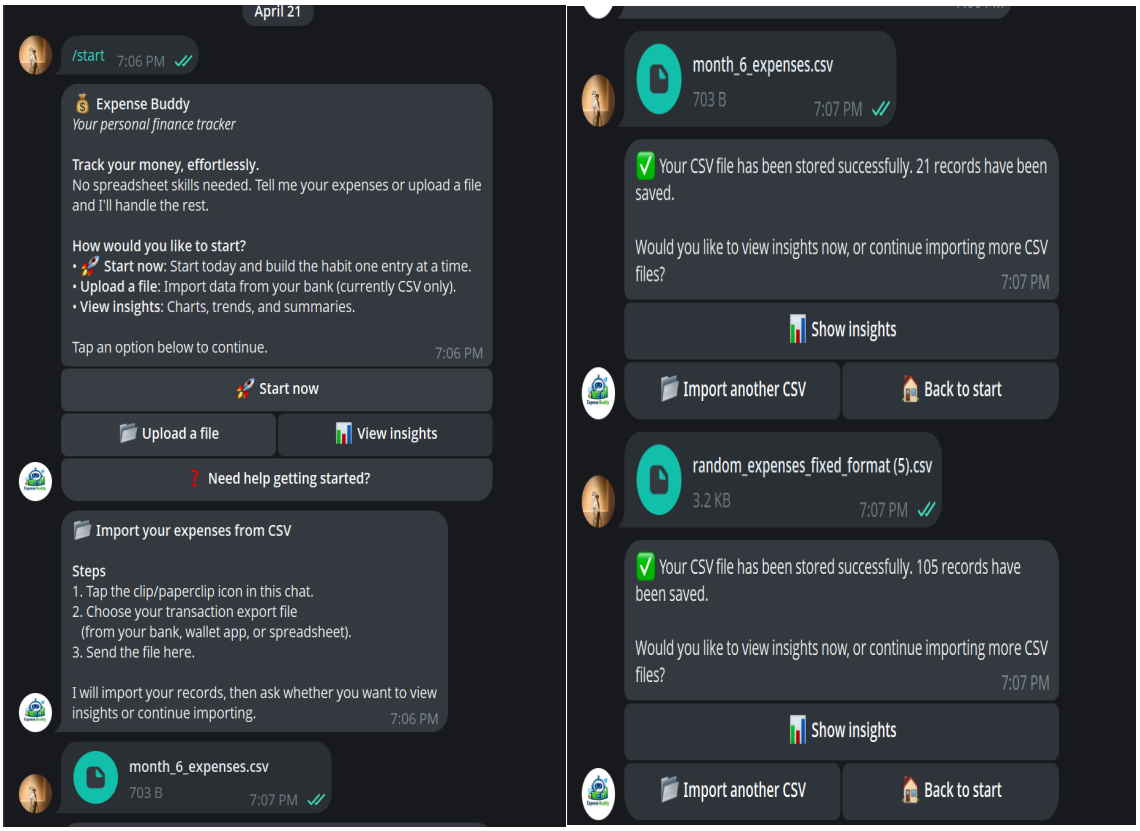
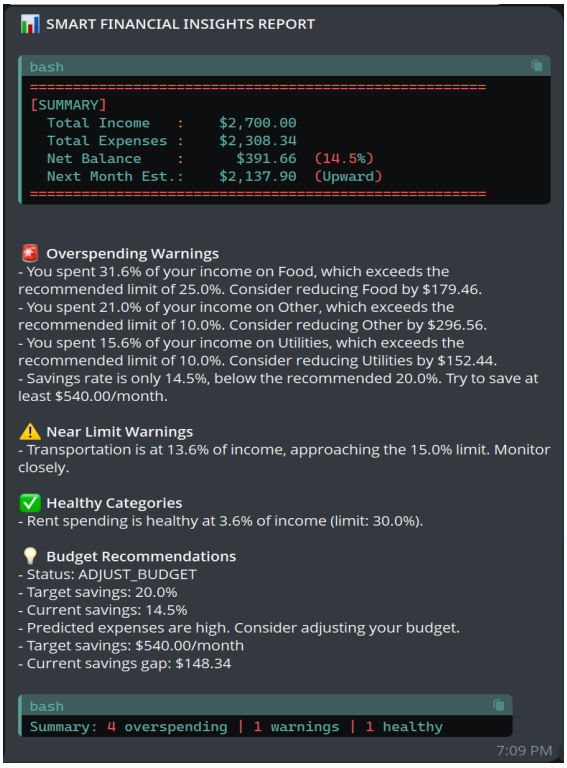


Figure 4 show the bot successfully identifying a transaction by interpreting the input text and extracting relevant information such as keywords and context. The system then maps the transaction to a predefined category, demonstrating its capability to handle unstructured data and perform accurate classification within the budgeting framework.

- Figure 5: Automated Budget Summaries and Visualizations and Predictive Insights and Rule-Based Financial Advice.



The financial summary output is as follows:

- Total Income: $2,700.00
- Total Expenses: $2,388.34
- Net Balance: $2,137.98 (14.5%)
- Next Month Estimate: $1,069.30 (Upward)
- Summary: 3 overspending | 1 warning | 1 healthy

Figure 5 shows the engine successfully detected Food (31.6%), Other (21.0%), and Utilities (15.6%) as exceeding their budget thresholds, identified Transportation (13.6%) as approaching its 15% limit, and generated specific dollar reduction recommendations for each violation. And show the full Personal Financial Dashboard generated by the pipeline displaying four panels: Income vs Expenses line chart with forecast marker, Spending by Category pie chart, Category Spending Over Time stacked bar chart, and Monthly Net Balance bar chart with the 20% savings target line.

VI. Discussion

The results of this project demonstrate that integrating multiple AI approaches into a single pipeline can effectively support personal financial analysis for students. Each model contributed differently to the overall system performance, revealing both strengths and limitations.

For Model 1 (Expense Classification), the TF-IDF + Logistic Regression approach achieved a very high accuracy of 99.86%, significantly outperforming the keyword-based baseline (78.71%). This result was expected, as machine learning models can capture contextual patterns in text beyond simple keyword matching. The inclusion of preprocessing steps such as text normalization and feature scaling further improved model performance by reducing noise in transaction descriptions. However, the extremely high accuracy may also be influenced by the structured nature of the dataset, which may not fully reflect real-world noisy inputs. Therefore, while the model performs well in controlled conditions, its performance may slightly decrease when applied to more diverse user-generated data.

For Model 2 (Spending Prediction), the Weighted Recent algorithm with a 2-month window produced the lowest MAE of $167.16. This indicates that short-term patterns are more useful than long-term trends for predicting personal expenses, especially when data is limited and irregular. The decision to replace Linear Regression with rolling-based methods proved effective, as Linear Regression failed to capture sudden spikes and seasonal variations. However, the prediction error remains relatively high, suggesting that the model still struggles with unexpected fluctuations in spending behavior. This limitation highlights the trade-off between model simplicity and predictive accuracy in small datasets.

For Model 3 (Rule-Based Financial Insights), the system successfully generated clear and interpretable financial recommendations. The rule-based approach proved effective in identifying overspending categories and providing actionable advice. Unlike black-box models, this method ensures transparency, which is essential for user trust. However, the

reliance on fixed thresholds limits flexibility, as different users may have different financial priorities or lifestyles. A more adaptive rule system could further improve personalization.

Despite the overall success of the system, several limitations were identified. First, the classification model depends on a pre-existing dataset that may not fully represent Cambodian spending habits. Second, the prediction model uses a relatively small dataset, which restricts its ability to generalize long-term trends. Third, the rule-based system applies static financial rules that may not suit all users. Additionally, the system was tested in a controlled environment using limited datasets, which may not reflect real-world deployment conditions.

In terms of scope, this project focuses on student-level financial analysis using lightweight and interpretable AI models. It does not include advanced deep learning methods or large-scale financial data integration. The system is designed primarily for demonstration and educational purposes rather than commercial deployment.

For future work, several improvements can be made. First, incorporating larger and more diverse datasets could improve both classification and prediction performance. Second, advanced time-series models such as LSTM or Prophet could be explored to better capture seasonal and nonlinear patterns. Third, the rule-based system could be enhanced with adaptive or learning-based mechanisms to provide more personalized recommendations. Finally, developing a full mobile or web application could improve usability and real-world impact.

VII. Conclusion

This project aimed to design and evaluate an AI-based budget tracking system that integrates multiple machine learning techniques to improve personal financial management for students. The motivation behind this work was to address the limitations of existing budgeting tools, which often lack automation, predictive capabilities, and meaningful insights.

To achieve this objective, a three-model pipeline was developed. Model 1 used a TF-IDF and Logistic Regression approach to automatically classify expense transactions. Model 2 applied rolling average and weighted recent algorithms to predict future spending based on historical data. Model 3 implemented a rule-based inference system to generate human-readable financial insights and detect overspending behavior.

The results demonstrate that the proposed system is effective in achieving its goals. Model 1 achieved a high classification accuracy of 99.86%, significantly outperforming the keyword-based baseline. Model 2 successfully identified short-term spending patterns, with the Weighted Recent algorithm providing the best performance. Model 3 generated clear and actionable financial recommendations, improving the interpretability of the system.

The key contribution of this project is the integration of multiple AI paradigms — supervised learning, time-series forecasting, and rule-based inference — into a single, lightweight, and interpretable system. This approach demonstrates how AI can be applied in a practical and accessible way to support better financial decision-making for students.

Overall, this project highlights the potential of combining simple yet effective AI techniques to create meaningful real-world applications. While there are limitations, the system provides a strong foundation for future improvements and further research in AI-driven personal finance tools.

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